

SOFTWARE SHOWCASE DOCUMENT

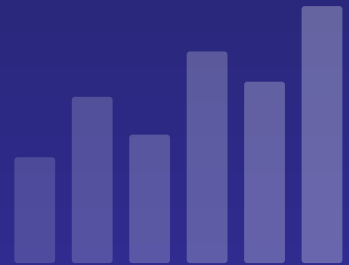
GymManager Lite

An end-to-end gym management system: members, memberships, sales, biometric access control, and cloud-based licensing.

Designed, architected and built end to end by

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ABOUT THE PROJECT

What is GymManager Lite

GymManager Lite is a desktop system for the day-to-day management of a gym: members, memberships, product sales with inventory control, configurable pricing, business KPIs, and biometric access control at the door. It ships as a self-contained graphical installer for Windows: the end user needs no Python, no database engine, and no internet connection to run it day to day.

The project was born as the streamlined version of **GymManager Pro**, a sibling product with real-time facial recognition, a dedicated member portal, and a Node/React stack. Lite keeps the operational core of the business and rebuilds it on a deliberately simpler architecture: Python and direct SQL, meant to be installed once on a front-desk computer and run autonomously for years, with updates governed by licensing.

Project at a glance

13	100%	~55 MB	11	128	7 days
functional modules	Python + SQL (no ORM)	self-contained installer	body measurements	face embedding values (no photos)	offline license grace period

ARCHITECTURE & ENGINEERING

Technical decisions and stack

Every layer was chosen deliberately to maximize simplicity, control, and portability in a local-installation context, with no infrastructure team behind it.

Layer	Technology	Why
Backend	Python 3 + Flask	A minimal framework with full control over the request cycle, no hidden magic.
Data	SQLite + direct SQL (no ORM)	Explicit parameterized queries: predictable performance and zero SQL injection surface.
Views	Jinja2 + hand-written CSS	Server-side rendering, no JavaScript frameworks, no build step.
Camera / photos	Targeted native JavaScript	The one exception: accessing the camera and live-cropping are browser capabilities with no server-side equivalent.
Biometrics	128-dimension facial embeddings	100% local comparison, no cloud, no stored images.
Licensing	Firebase (Firestore + anonymous Auth) over REST	A managed backend with declarative security rules, no server of its own to maintain.
Packaging	PyInstaller + graphical installer (Tkinter)	A single self-contained .exe: no Python, no dependencies on the client machine.

Design principles applied

- **The database as a single, portable file:** the whole gym lives in one SQLite file, with schedulable automatic backups and validated restores (integrity check, backs up the current state, and asks for a password before replacing data).
- **Domain-driven modular architecture:** one view file per business module (members, sales, pricing, users, audit log, backups, licensing...), which keeps maintenance and extension straightforward.
- **License / data separation:** the license deliberately lives outside the gym's database, so a backup restored on another machine never drags along someone else's activation credentials.
- **No persistent background timers:** license resynchronization is evaluated lazily on each protected request, avoiding unnecessary background threads.

FUNCTIONALITY

System modules

Thirteen modules cover the full operating cycle of a gym, from a member's first sign-up to closing the cash register and auditing everything that happened in the system.

Module	What it solves
Home	Live KPIs: overdue and soon-to-expire memberships, 12-month charts, monthly revenue, best sellers, and low stock.
Member records	Sign-up, search, filters, and editing, with a photo taken from the webcam and a built-in circular-crop editor.
Gym enrollment	Board filterable by status (active / overdue / no membership), monthly plans with add-on services, pause and resume, printable receipt.
Product sales	Catalog, cart, buyer autocomplete, real-time stock control, receipt with barcode, and history.
Product catalog	Create, edit, add images, and soft-delete products.
Pricing	Enrollment prices by duration, additional-month rate, and add-on services, calculated by quantity.
Refunds	Full or partial refunds by barcode, with refundable balance, mandatory reason, and net reporting in revenue.
Fitness profile	Eleven body measurements with automatic calculation of BMI, body fat, waist-to-hip ratio, and waist-to-height ratio, with configurable interpretation ranges.
Data migration	Bulk import from Excel with row-by-row validation before saving, for clients switching from a previous system.
Users and roles	Admin, Front Desk, and Trainer, each with its own screen scope.
Activity log	Action audit trail filterable by role, user, and date, exportable to CSV.
Backups	Schedulable, with configurable retention and validated restore from the interface itself.
Access control	Facial-recognition kiosk at the door, with an entry history (see next page).

FEATURE SPOTLIGHT

Biometric access control

The access kiosk turns any front-desk computer with a webcam into a real-time entry checkpoint: it recognizes the member's face and decides whether they may enter, with no staff intervention.

How it works

- **1. Guided enrollment:** during sign-up, 5 face samples are captured with on-screen prompts (facing forward, turning, chin up), each one accepted automatically once the framing is correct.
- **2. Recognition at the door:** with the kiosk tab open, every person who steps in front of the camera is identified and checked against their membership status within seconds.
- **3. Clear decision:** access granted (showing photo, ID, expiration date, and phone number), access denied for an expired or missing membership, face not recognized, or a prompt to step closer.
- **4. Configurable debounce:** someone standing in front of the camera does not generate repeated entries; the lockout window ranges from 90 seconds up to 12 hours depending on the gym's policy.
- **5. Live, non-intrusive alerts:** while staff work on another screen, a discreet notification shows who just walked in, even when several people arrive in a row.

Privacy by design. No camera image is ever stored. Each face is reduced to 128 numeric values that describe it mathematically -the face cannot be reconstructed from them- and the comparison happens entirely on the local machine: no biometric data ever leaves the device.

Other business features

- **Thermal receipts:** tickets for 58/80 mm printers, with configurable business details and a true-to-size preview before printing.
- **Traceable refunds:** by barcode, partial or full, with reason, payment method, and the staff member responsible, never altering the original receipt.
- **Pause a membership:** freezes a member's remaining days (travel, injury) and restores them intact on resume, without flagging them as overdue in the meantime.
- **Assisted migration:** an Excel template with row-by-row validation up front, for bringing in a member base from a previous system without losing historical revenue.

FEATURE SPOTLIGHT

Cloud-based licensing

Although the application runs 100% locally, each license's activation and validity are checked against a cloud backend, which enables a multi-tenant commercial model on top of a product installed on every gym's own machine.

Element	Design
Tiers	Trial, monthly, annual, and perpetual, each gym with its own independent license.
Device identity	A unique hardware fingerprint (machine ID + disk serial + computer name), to prevent sharing one license across multiple machines.
Backend	Firebase Firestore, with anonymous authentication and declarative security rules that only allow reading one's own license, never listing others.
Offline mode	A local cache signed with HMAC-SHA256 and encrypted with Windows DPAPI, with a 7-day grace period if the machine loses internet access.
Trusted clock	Detects system-clock tampering, to prevent bypassing expiration by rolling the clock back.
Vendor tooling	A dedicated CLI to issue, renew, revoke, reactivate, and release licenses, with no self-service inside the app itself.

Why it lives outside the gym's database

The license is deliberately stored in files separate from `gym.db`. The backup module copies that file as-is, so any activation credential living inside it would travel with every backup restored on machines that should never have it. Keeping it separate closes off that leak through a path that was never meant for it.

QUALITY & OPERATIONS

Security and deployment

Security

- Passwords hashed with **bcrypt**, never stored in plain text.
- Account lockout for 5 minutes after 3 failed login attempts.
- Signed, **HttpOnly** session cookie, expiring after 8 hours.
- **CSRF** protection on every form in the application.
- SQL queries always use **bound parameters**: zero injection surface.
- Deleting a member or a user requires re-entering the password of the currently logged-in session.
- Uploaded images are saved with a random name and a validated extension.
- License cache protected with Windows **DPAPI** encryption when available.

Installation and distribution

The entire packaging process resolves to a single command that compiles the application, the uninstaller, and the graphical wizard, producing a self-contained Windows installer of roughly 55 MB.

- A custom graphical wizard (license agreement, destination folder, progress, shortcuts).
- Detects previous installations: when updating, it asks whether to keep the existing data.
- A clean database with a single administrator user once installation finishes.
- An uninstaller that preserves gym data unless explicitly told to delete it.
- Requires no Python, no database engine, and no administrator rights on the client machine.

Thank you for reading

This document summarizes the design, architecture, and development of GymManager Lite, built end to end as an independent project.

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Open to new opportunities and collaborations